Temporal Measurement Instability in LLM Benchmarks: A Large-Scale Psychometric Audit of 5,227 Models

Anonymous ACL submission

Abstract

LLM benchmark scores are compared across model generations, yet whether individual items measure the same construct over time is rarely tested. Existing contamination detectors validate against web-overlap labels that conflate global exposure with differential functioning. We apply differential item functioning (DIF) analysis from educational measurement to six METABENCH benchmarks (5,227 MMLU models, 27,125 items across all six), validated through a semi-synthetic injection framework with cross-subject matching. Between 2023 and 2024 model cohorts, 17.9–31.3% of items exhibit large temporal DIF; for MMLU, two convergent non-independence corrections yield approximately 27–29% (conservative bound: 13.5%; cluster bootstrap 95% CI $[26.7\%, 53.2\%]$; random-split baseline 0.18%). Web-overlap contamination labels are near-independent of DIF flags (Jaccard $= 0.206$): the two constructs—global exposure and differential functioning—are empirically orthogonal. This instability does not threaten aggregate rankings ($\rho \approx 0.997$) but reveals item-level temporal change, with 2024 models gaining on hard items while eroding on easy ones. LLM benchmarks need psychometric monitoring, not just score reporting.

1 Introduction

When we say "Model B outperforms Model A on MMLU," we implicitly assume that the benchmark measures the same construct in the same way for both models. This assumption of temporal comparability underpins every cross-generation comparison published on leaderboards and in model reports, yet it is rarely tested. We use *measurement invariance* to denote the property that test items function identically across groups (Meredith, 1993), *differential item functioning* (DIF) as the item-level statistical test for violations, and *temporal measurement instability* as the empirical phenomenon when DIF is detected across time-defined cohorts—a descriptive term denoting systematic changes in measurement properties, without implying degradation or attributing cause. Applying Mantel-Haenszel DIF analysis (Holland and Thayer, 1988) with ETS severity classification to the METABENCH response matrix (Kipnis et al., 2024),[1] we find that 31% of MMLU (Hendrycks et al., 2021) items receive ETS Category C classification ($|\Delta_{\mathrm{MH}}| \geq 1.5$, $p < 0.05$) between 2023 and 2024 model cohorts—versus 0.18% under a random-split control. This instability does not threaten benchmark validity ($\rho \approx 0.997$ between full and stable-item rankings); it is diagnostic, revealing which capability dimensions shift across model generations.

Contamination detectors—from membership inference (Shi et al., 2024) to performance-based methods (Dekoninck et al., 2024; Liang, 2026; Tao et al., 2026)—validate primarily against web-overlap labels (Li et al., 2024b) that measure *global exposure*: whether an item exists in publicly accessible training corpora. Such labels do not capture *differential item functioning*: whether an item behaves differently across cohorts with varying exposure or training methodology (Figure 1). Direct comparison confirms near-independence (Jaccard $= 0.206$, $\phi = 0.012$), consistent across contamination subtypes.

We apply a temporal DIF audit framework to the full METABENCH response matrix (5,227 models $\times$ 12,508 MMLU items), to our knowledge the largest-scale DIF analysis on LLM benchmarks, complementing concurrent work on anchor-item calibration (Habba et al., 2026) and addressing the fragility of existing detectors under post-training perturbation (Wang et al., 2026). Two structural controls, a sensitivity check, and replication across all six METABENCH benchmarks

[1] Analysis code and DIF audit scripts are available at https://anonymous.4open.science/r/dif-audit.

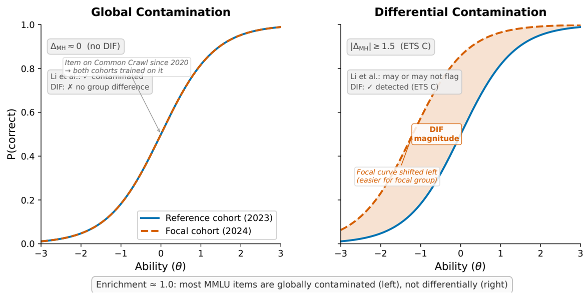

Figure 1: Schematic illustration of global vs. differential contamination via Item Characteristic Curves (ICCs). **Left**: global contamination produces no DIF; both cohorts encounter the item, and their ICCs overlap. **Right**: differential contamination produces DIF; the focal cohort's ICC shifts because only that cohort was disproportionately exposed during training. The enrichment ratio of ≈ 1.0 between DIF flags and web-overlap labels is expected because most MMLU items fall into the left category.

confirm that the temporal DIF rate (≈27–29% after non-independence corrections) is not a methodological artifact (Sections 4.2, 4.4 and 4.7).

DIF should be understood as a diagnostic lens rather than a quality filter: DIF-flagged items concentrate discriminative signal ($z = -8$ to -49 vs. random removal; Section 4.6), and model families show distinctive DIF profiles.

Our contributions are methodological and empirical, not algorithmic:

1. **Orthogonal constructs**: We empirically demonstrate that web-overlap labels (Li et al., 2024b), which index global exposure, are near-independent of DIF flags, which index differential functioning (Jaccard $= 0.206$, $\phi = 0.012$ on 9,229 items); the two capture fundamentally different signals (Section 4.3).

2. **Temporal DIF audit framework**: We develop and validate a general-purpose audit framework for multi-domain benchmarks, comprising cross-subject matching (ΔFPR $= 0$ guarantee), family-level clustering, per-item design-effect correction, and base-vs-instruct mechanism decomposition; validated semi-synthetically and replicated on GSM8K (Sections 4.5 and 4.7).

3. **Temporal measurement instability at scale**: We establish that ≈27–29% of MMLU items exhibit temporal DIF after non-independence corrections; replication across six METABENCH benchmarks yields C% = 17.9–31.3%. Base-vs-instruct stratification reveals mechanistically distinct patterns ($\chi^2 = 562.7$), providing positive evidence for mechanism separation (Sections 4.2, 4.7 and 5.1).

2 Related Work

Contamination Detection. A growing family of methods detects benchmark contamination through model behavior or training data analysis. Membership inference approaches estimate whether a model has seen specific items during training: Min-K% (Shi et al., 2024) and Min-K%++ (Zhang et al., 2024) use token-level log-likelihood statistics, while Time Travel (Golchin and Surdeanu, 2024) probes models with temporal cues and Oren et al. (2024) prove contamination through exchangeability tests. Performance-based detectors identify contamination through statistical anomalies (Dekoninck et al., 2024; Liang, 2026; Tao et al., 2026; Deng et al., 2024). Balloccu et al. (2024) provide a broader taxonomy of leakage detection strategies. These methods share a common validation practice: they verify detection accuracy against web-overlap or n-gram labels (Li et al., 2024b) that classify items based on verbatim matches in publicly accessible corpora. Such labels measure *global exposure* (whether an item exists in training data) rather than *differential item functioning* (whether an item behaves differently across model cohorts). This global-vs-differential distinction motivates our ground truth analysis (Section 4.3). Work on LLM memorization (Carlini et al., 2023; Tirumala et al., 2022) establishes that memorization depends on data frequency and model scale, reinforcing the distinction between global exposure and differential behavior. Wang et al. (2026) show that existing contamination detectors are fragile under reinforcement learning, further motivating behavioral approaches grounded in psychometric theory.

IRT in LLM Evaluation. IRT adoption in LLM evaluation has accelerated: Lalor et al. (2019) introduced IRT to NLP, Vania et al. (2021) assessed test set discriminability, and Rodriguez et al. (2021) demonstrated that not all evaluation examples are equally informative. Recent extensions include PSN-IRT (Zhou et al., 2026) for benchmark quality at scale, TinyBenchmarks (Maia Polo et al., 2024) for IRT-based item selection, scalability coefficients for detecting bad benchmark items

(Hardy et al., 2026), IRT-based evaluation quality (Kraus and Arora, 2024), ATLAS (Li et al., 2025) for computerized adaptive testing of LLMs, and competency-focused IRT evaluation in medical domains (Luo et al., 2025). The closest concurrent work is Growing Pains (Habba et al., 2026), which uses IRT calibration to establish anchor items for cross-temporal score equating via fixed-parameter calibration (FPC; Kim and Kolen, 2019). While Growing Pains prescribes how to equate scores after instability is detected, our work diagnoses *which* items exhibit instability and quantifies its magnitude—the diagnostic step that must precede equating. Their anchor-item selection can directly consume our stable-item set as pre-screened candidates; our finding that 31% of items exhibit temporal DIF identifies which items should *not* serve as anchors. To our knowledge, no prior work applies DIF to benchmark temporal comparability—contamination detection and psychometric evaluation optimization have developed as separate threads.

DIF Methodology and Measurement Invariance. The Mantel-Haenszel procedure for DIF detection was introduced by Holland and Thayer (1988) and remains the most widely used method in educational testing due to its simplicity and well-characterized statistical properties (Clauser and Mazor, 1998). Alternative approaches include logistic regression DIF (Swaminathan and Rogers, 1990), SIBTEST (Shealy and Stout, 1993), Lord's χ^2 test (Lord, 1980), and the Raju area method (Raju, 1988); Millsap and Everson (1993), Zumbo (1999), and Penfield and Camilli (2007) provide comprehensive reviews. DIF analysis is a specific instance of measurement invariance testing (Meredith, 1993; Vandenberg and Lance, 2000; Millsap, 2011), which examines whether a measurement instrument functions equivalently across populations. MIMIC models offer a latent-variable approach to the same problem (Muthén, 1985; Woods, 2009), and Teresi and Fleishman (2007) reviews DIF methods across applied domains. Cross-subject matching—computing the stratification variable from domains other than the one being tested—has psychometric precedent in the DIF literature (Nandakumar, 1993; Donoghue and Allen, 1993); our adaptation addresses LLM-specific challenges of multi-domain local dependence and simultaneous contamination control at scale. We do not modify the MH procedure; our contribution is the application context and the empirical findings it reveals. The 5,227-examinee scale exceeds typical DIF applications (Belzak, 2020). The impact-versus-DIF confound is addressed through matched-ability quintile analysis, IRT θ-conditioning (as sensitivity check), and a random-split control (Section 4.2).

Benchmark Quality and Design. Several concurrent approaches address benchmark reliability from complementary angles. The Benchmark Health Index (Zhu et al., 2026) and Benchmark^2 (Qian and Huang, 2026) perform meta-evaluation of static benchmark quality at a single snapshot; our temporal DIF analysis diagnoses *when* measurement properties shift. MMLU-CF (Zhao et al., 2025) reconstructs contamination-free items, and MMLU-Pro (Wang et al., 2024) extends item difficulty; DIF monitoring provides a lower-cost alternative when reconstruction is infeasible, identifying which existing items remain measurement-invariant rather than replacing them. Combined with the prescriptive equating framework of Growing Pains (Habba et al., 2026) (discussed above), these approaches span the full lifecycle from diagnosis (our work) to reconstruction (Zhao et al., 2025) to calibration (Habba et al., 2026). Maeda (2025) predict which item features associate with DIF in educational settings; we instead detect DIF empirically across LLM cohorts at scale. Dynamic benchmarks (Kiela et al., 2021; Li et al., 2024a; White, 2025) and self-evolving update frameworks (Ying et al., 2024; Xu et al., 2025) mitigate contamination by construction, yet none provides psychometric equating across versions. Leaderboard rankings are sensitive to minor evaluation perturbations (Alzahrani et al., 2024). Broader surveys (Chang et al., 2024; Burnell et al., 2023) identify benchmark saturation and item-level opacity as systemic challenges that our DIF analysis directly addresses.

3 Method

We adopt Differential Item Functioning (DIF) analysis from educational measurement to assess whether individual benchmark items behave consistently across groups of LLMs. The procedure requires only a binary response matrix and group labels, with no access to model internals or training data. Below we define the data, the statistical test, the cohort designs, and a semi-synthetic validation framework.

3.1 Data

The primary dataset is the METABENCH response matrix (Kipnis et al., 2024), which records binary outcomes (1 = correct, 0 = incorrect) for 5,227 models evaluated on 12,508 MMLU items (Hendrycks et al., 2021). Models span release dates from 2023 to 2024, drawn from the Open LLM Leaderboard. MMLU covers 57 subjects grouped into five broad domains (STEM, Humanities, Social Sciences, Other, and a mixed category). We supplement the response matrix with the web-overlap contamination labels of Li et al. (2024b), which classify 9,229 of 12,508 items (73.8% coverage) as "contaminated" or "clean" based on verbatim matches in publicly accessible web corpora. These labels serve as an external reference for evaluating the relationship between DIF flags and existing contamination indicators.

3.2 Mantel-Haenszel DIF and ETS Classification

In educational testing, Differential Item Functioning (DIF) occurs when an item's statistical relationship to the latent trait θ differs between two groups after conditioning on ability level. An item with DIF is not merely harder for one group; it functions differently relative to what the group's overall ability would predict. The standard terminology refers to test-takers as "examinees" and questions as "items." In our setting, examinees are LLMs and items are benchmark questions.

The Mantel-Haenszel (MH) procedure (Holland and Thayer, 1988; Magis et al., 2010) detects DIF by stratifying examinees into K score bins based on total score (a proxy for θ) and computing a common odds ratio across strata. We use $K = 5$ equiprobable strata (quintile-based binning) following ETS recommended practice for large-scale assessments (Donoghue and Allen, 1993; Clauser and Mazor, 1998). Within each stratum k, the responses of the reference group and the focal group form a 2×2 contingency table:

$$\alpha_{\mathrm{MH}} = \frac{\sum_{k=1}^{K} A_k D_k \,/\, T_k}{\sum_{k=1}^{K} B_k C_k \,/\, T_k}, \qquad (1)$$

where A_k and B_k are the number of correct and incorrect responses in the reference group for stratum k, C_k and D_k are the corresponding counts for the focal group, and T_k is the total number of examinees in stratum k. Under the null hypothesis of no DIF, $\alpha_{\mathrm{MH}} = 1$. The log odds ratio is converted to the ETS delta scale:

$$\Delta_{\mathrm{MH}} = -2.35 \ln(\alpha_{\mathrm{MH}}) \,. \qquad (2)$$

The Educational Testing Service (ETS) classifies DIF severity into three categories based on $|\Delta_{\mathrm{MH}}|$. Category A (negligible) requires $|\Delta_{\mathrm{MH}}| < 1.0$. Category C (large) requires both $|\Delta_{\mathrm{MH}}| \geq 1.5$ and statistical significance at $p < 0.05$ from the MH chi-squared test. Category B (moderate) includes all remaining items: those with $1.0 \leq |\Delta_{\mathrm{MH}}| < 1.5$, or $|\Delta_{\mathrm{MH}}| \geq 1.5$ without statistical significance. The dual threshold in ETS C (effect size *and* significance) guards against flagging items that show large but unreliable effect sizes. Throughout this paper, we report the percentage of items classified as ETS C, denoted C%. At $N = 5{,}227$, significance ($p < 0.05$) is nearly automatically satisfied for any $|\Delta_{\mathrm{MH}}| \geq 1.5$: only 3 of 3,918 C items lack significance, so the dual criterion effectively reduces to a single effect-size threshold; per-item DEFF correction (Section C) restores p-value selectivity by reducing effective N: under simple-mean DEFF, 269 items (6.9%) lose significance while retaining $|\Delta_{\mathrm{MH}}| \geq 1.5$; 74% of these are near-miss ($p_{\mathrm{adj}} \in [0.05, 0.10]$).

The choice of matching variable (the total score used for stratification) affects both validity and interpretation. We distinguish three matching strategies: (1) *standard matching* uses the within-benchmark total score on all items of the same benchmark; (2) *cross-subject matching*, adapting purified subtest matching (Swaminathan and Rogers, 1990) to the LLM multi-subject setting, uses the total score on all subjects *other than* the one containing the item under test, preventing the matching variable from absorbing injected contamination signals (Section 3.4); and (3) *external matching* uses the total score on a different benchmark entirely (e.g., MMLU total score as the matching variable when testing DIF on GSM8K items), providing an independent ability proxy uncontaminated by the target benchmark. Cross-subject and external matching both address matching variable contamination but at different granularities: cross-subject operates within a multi-domain benchmark, while external operates across benchmarks. Total score matching is sufficient: IRT θ correlates at $r = 0.9986$ with total scores (Section 4.4).

LLM-specific challenges. Applying MH-DIF to LLMs introduces assumptions absent in educational testing. First, overlapping training corpora violate local independence; permutation simulation confirms LD-attributable FPR inflation < 0.001 pp under cross-subject matching (Section A). Second, fine-tuned variants and merges within model families are not independent examinees; graduated deduplication shows family structure does not inflate C% (Section 4.4). Third, bimodal ability distributions (base vs. instruction-tuned) motivate quintile-based rather than parametric stratification. Fourth, MMLU's 57-subject structure raises a multidimensionality concern: if 2023 and 2024 cohorts differ on secondary ability dimensions, total-score matching may produce spurious DIF (Ackerman, 1992; Roussos and Stout, 1996). We address this through three lines of evidence: (i) domain-level ability profiles are nearly flat across cohorts (Cohen's d: STEM $= 0.26$, Humanities $= 0.16$, Social Science $= 0.15$, Other $= 0.15$; range $= 0.12$), indicating uniform improvement that total-score matching adequately captures; (ii) cross-subject and within-subject matching show strong agreement ($\kappa = 0.80$, 5 subjects, 834 items; the 20% discordance is distributed across domains rather than concentrated in specific subjects), with cross-subject matching yielding 2.7 pp more C items—an upper bound on any multidimensional confound; (iii) cross-subject-unique C items do not concentrate in subjects with large cohort ability shifts, further dissociating multidimensional confound from the observed DIF signal (Sections B and 4.4). Unlike standard purified MH that conditions on total test score, we match on cross-subject total score to avoid criterion contamination when target-subject items themselves may be compromised—an adaptation specific to multi-domain LLM benchmarks where subjects may be differentially affected.

3.3 Cohort Designs

We construct five cohort comparisons: (1) *Temporal*: 2023 ($N = 1{,}080$) vs. 2024 ($N = 807$) models, the primary comparison; (2) *Ability-quintile*: Q1–Q2 vs. Q4–Q5 within the same temporal window; (3) *Within-family*: Llama-2 ($N = 161$) vs. Llama-3 ($N = 308$), isolating generational changes within a single family; (4) *Within-subject*: per-subject DIF to test domain concentration; and (5) *Random split*: two random halves of the 2023 cohort ($N = 540$ each) as a sanity check.

3.4 Semi-Synthetic Injection Framework

To evaluate whether MH-DIF can detect differential contamination when it is present, we construct a semi-synthetic framework with controlled ground truth.

Procedure. From items classified as ETS A in the uninjected data (negligible DIF), we select 20% uniformly at random as the "contaminated" set. For each contaminated item, we flip incorrect responses to correct in the focal group at exploitation rate $p \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$. We then run MH-DIF on the modified response matrix and measure detection performance. We report $\Delta\text{TPR} = \text{TPR}(\text{injected}) - \text{TPR}(\text{baseline})$ and $\Delta\text{FPR} = \text{FPR}(\text{injected}) - \text{FPR}(\text{baseline})$, where TPR is the true positive rate on injected items and FPR is the false positive rate on uninjected items. Each injection level is repeated across 5 random seeds for uncertainty estimation. Because binary response flipping represents a worst-case uniform injection pattern, reported ΔTPR values constitute upper-bound estimates; realistic contamination (partial memorization, format-specific advantages) would produce weaker signals.

Cross-subject matching. The choice of matching variable is critical for valid semi-synthetic evaluation. When contamination is injected into items within subject X, the total score on subject X absorbs part of the injection signal. Using this contaminated total score as the MH matching variable violates the conditional independence assumption and inflates FPR. Cross-subject matching resolves this by computing the matching variable from all subjects *other than* the one being tested.

Remark (Zero FPR under single-subject matching). Let $M_i^{(-s)} = \sum_{j \notin s} Y_{ij}$ be the matching variable for examinee i on focal subject s. Under single-subject injection (contamination confined to subject s), the tested item belongs to s and does not contribute to $M_i^{(-s)}$; injecting contamination into s leaves $M_i^{(-s)}$ unchanged. Therefore $\Delta\text{FPR} = 0$ by construction.

When contamination spans multiple subjects simultaneously, $M_i^{(-s)}$ may absorb signal from other contaminated subjects, potentially inflating FPR. Empirically, this guarantee extends to multi-domain contamination affecting $\leq$9% of subjects ($k \leq 5$), with graceful degradation beyond this boundary (Section 4.5).

Empirically, ΔFPR $= 0.000$ across all five MMLU domains (bootstrap 95% CI $= [0.000, 0.000]$). ΔFPR $= 0$ is a constructive guarantee of the matching strategy, independent of injection realism, whereas ΔTPR constitutes an upper-bound estimate under uniform injection. ΔFPR remains negligible for single-domain contamination ($k \leq 5$); when contamination spans many domains, within-subject matching should be preferred (Section 4.5). We adopt cross-subject matching as the standard protocol.

Following DIF convention (Wainer and Braun, 1993), we do not apply family-wise error correction: the dual ETS C threshold ($|\Delta_{\mathrm{MH}}| \geq 1.5$ *and* $p < 0.05$) guards against false positives; BH-FDR at $q = 0.05$ reclassifies only 3 of 3,918 C items (Section A). Diagnostic procedures (Yen's Q3, domain-specificity, DIF decomposition) and further details appear in Section A.

4 Experiments

We apply MH-DIF to the METABENCH response matrix (Kipnis et al., 2024) (5,227 models $\times$ 12,508 MMLU items (Hendrycks et al., 2021)), with temporal cohorts defined by release date (2023 vs. 2024). Cross-benchmark validation across all six METABENCH benchmarks appears in Section 4.7.

4.1 Validation: DIF Detection Power

As approximate parametric bounds, a Monte Carlo power analysis (1,000 replications per condition) using empirical 2PL parameters from PSN-IRT (Zhou et al., 2026) ($\bar{a} = 1.2$, $\bar{b} = -0.84$; 14,042 MMLU items) confirms that at $N \geq 80$ per cohort, ETS C achieves TPR of 0.68–0.80, FPR of 0.036–0.042, and AUC of 0.93–0.94 for items with $|\Delta_{\mathrm{MH}}| \geq 1.4$ (details in Section A). Our temporal cohorts ($N_{\mathrm{ref}} = 1{,}080$, $N_{\mathrm{focal}} = 807$) far exceed this threshold. Power varies substantially with item discrimination: TPR ranges from 0.40 at $a = 0.5$ to 0.95 at $a = 2.0$; the gap between TPR at mean a (0.87) and the aggregate TPR (0.68) quantifies the power cost of parameter heterogeneity, consistent with Jensen's inequality (Section A.2).

4.2 Temporal Comparability: 31% Measurement Instability

We compare 2023 ($N = 1{,}080$) against 2024 ($N = 807$) models; 31.3% of items receive ETS C classification ($|\Delta_{\mathrm{MH}}| \geq 1.5$; 95% bootstrap CI: [29.4%, 38.1%]; family-level cluster bootstrap CI $[26.7\%, 53.2\%]$). Two independent non-independence corrections converge on approximately 27–29%: simple-mean DEFF correction (29.2%) and family-level cluster bootstrap lower bound (26.7%). The weighted-mean DEFF variant yields 13.5%, but its effective cluster size ($\bar{m}$=254) implies each family member is functionally identical—unrealistic: within-family similarity is flat across family sizes ($\rho_s = -0.009$, $p = 0.96$; Section H), so the weighted variant over-corrects. The $[13.5\%, 29.2\%]$ range reflects power uncertainty, not DIF magnitude disagreement ($|\Delta_{\mathrm{MH}}| \geq 1.5$ unchanged). We adopt $\approx$27–29% as our primary estimate, with 13.5% as a lower bound under weighted-mean DEFF (Sections C and D); even this bound exceeds the random-split baseline by 75$\times$. Leave-one-family-out analysis confirms no single family drives the signal ($|\Delta\mathrm{C}\%| \leq 2.93$ pp). Graduated de-duplication independently confirms C% scales with power, not family redundancy: at each fixed K, within-K SD $<$1.5 pp across random draws (Section I). C% is stable across $K \geq 5$ (31.3%–35.9%; K=5 most conservative); the non-monotonic K=10 peak reflects ability–DIF interactions in finer strata, consistent with residual non-uniform DIF that MH does not capture, while K=3 inflates to 39.4% from under-stratification.

Three controls confirm genuine temporal change (Table 1): random splits yield C% of 0.18% (2023) and 1.3% (2024), both $\geq$23$\times$ below temporal C%; ability-quintile comparison yields 2.9%; and 1:1 nearest-neighbor matching on total score (N=638/group, Cohen's $d = -0.17$) yields C% $= 28.9\% \pm 0.1\%$, confirming at most $\sim$2.4 pp is attributable to ability mismatch ($p < 0.01$; Section A).

Across all 57 subjects, C% ranges from 20.7% to 48.4% (mean $\approx$ 30%; Table 1). DIF rates are domain-invariant (Kruskal-Wallis $p = 0.78$), but DIF *direction* exhibits domain dependence: STEM items favor 2024 models (55.4%), Humanities favor 2023 (64.4%). Domain-level accuracy gains explain $<$0.1% of item-level DIF variance (pseudo $R^2 = 0.0007$); a Simpson's paradox confirms DIF is associated with item properties rather than aggregate gains (OR reverses from 1.55 to 0.73 after controlling for difficulty; Section B). A difficulty–direction interaction emerges ($\rho = -0.597$): 2024 models gain on hard items while eroding on easy ones. The conclusion is robust across thresholds (18.3% at $|\Delta_{\mathrm{MH}}| \geq 2.0$, 10.2% at ≥ 2.5).

Table 1: DIF landscape across cohort definitions. ETS C denotes the percentage of items with $|\Delta_{\text{MH}}| \geq 1.5$. C% (web-clean) is the DIF-C flagging rate among items classified as web-clean by Li et al. (2024b). Enrichment is the ratio of observed to expected overlap between DIF flags and web-overlap labels. Dashes (–) indicate comparisons where Li et al. labels are not applicable.

Cohort Definition	N_{ref} / N_{focal}	C%	C% (web-clean)	Enrich.
Temporal (2023 vs. 2024)	1,080 / 807	31.3	0.318	0.964
Ability quintile (Q1–2 vs. Q4–5)	∼2,600 / ∼2,600	2.9	0.027	1.19
Within-family (Llama-2 vs. 3)	161 / 308	61.3	–	–
Random split 2023 (sanity)	540 / 540	0.18	–	–
Random split 2024 (sanity)	∼400 / ∼400	1.3	–	–
Within-subject (avg, 5 subj.)	*varies*	38.1	–	–

4.3 Ground Truth Mismatch

Direct comparison of DIF flags against the web-overlap labels of Li et al. (2024b) on 9,229 annotated items confirms that the two methods capture orthogonal signals. The enrichment ratio is 0.964 (bootstrap 95% CI: $[0.94, 1.02]$; Table 1): DIF-C rates are 30.7% among web-contaminated items and 31.8% among clean items. Association tests confirm near-independence ($\phi = 0.012$, OR $=$ 1.05 [0.96, 1.15], Jaccard $=$ 0.206, within permutation null $[0.203, 0.222]$; $p = 0.88$). This near-independence holds across contamination subtypes. The null result does not indicate detection failure: semi-synthetic injection produces a clear dose-response (ΔTPR $= +0.508$ at $p = 0.5$; Figure 2). The disconnect arises because web-overlap labels measure *global exposure* (whether an item exists in training corpora), while DIF measures *differential functioning* (whether an item's relationship to ability changes across cohorts). The two constructs—global exposure and differential functioning—are empirically orthogonal.

4.4 Structural Analysis

Two structural controls and a sensitivity check fail to reduce temporal C% below 31% (Table 2): matched-ability quintiles, within-subject DIF (C% $=$ 38.1%; cross-subject C% is more conservative due to lower local dependence), and IRT θ-conditioning ($r(\theta, \text{total score}) = 0.9986$). Logistic regression on nine subjects (1,787 items; 9/57—full-benchmark rates require further analysis) detects an additional 17.7% with non-uniform DIF ($|\beta_{\text{interaction}}| \geq 0.5$; effect-size criterion; LR p-values not adjusted for clustering); the union reaches 47% (Jaccard $=$ 0.28): ≈27–29% uniform (MH) plus 17.7% non-uniform (interaction-only), mutually exclusive by construction. Ability-matched analysis yields

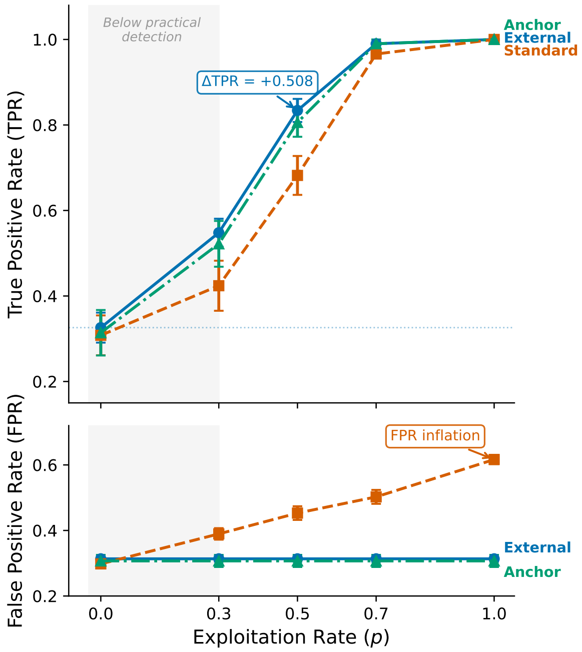

Figure 2: Semi-synthetic dose-response for DIF detection on MMLU. ΔTPR increases monotonically with exploitation rate p, confirming that MH-DIF is sensitive to differential item behavior when present. Baseline (no injection) produces $\Delta\text{TPR} \approx 0$.

Table 2: Structural controls and sensitivity checks for temporal DIF. None reduces C% below 31%; θ-conditioning serves as a sensitivity check confirming total-score adequacy.

Approach	Mechanism	Temp. C%	TPR (p=0.5)	Outcome
Baseline (ext.)	Cross-bench. total score	$0.314_{\pm.009}$	$0.834_{\pm.027}$	Reference
Matched-ability	Within-quintile temp. DIF	∼0.31	–	Enrich. 0.94–1.04
θ-cond. (sens.)	θ replaces total score	0.312	0.876	$r(\theta, \text{total})$=0.9986
Within-subject	Per-subject DIF	–	–	C% $=$ 38.1% (>25%)

C% $=$ 28.9% (temporal) and 28.3% (Llama-2 vs. 3)—consistent matched values (∼28–29%) despite vastly different unmatched rates (Section B).

4.5 Semi-Synthetic Framework Validation

Cross-subject matching achieves $\Delta\text{FPR} = 0.000$ across all five MMLU subjects (4/5 with $\Delta\text{TPR} > 0.3$ at $p = 0.5$, upper-bound estimate under uniform binary flipping; Section 3.4), while within-subject matching inflates FPR by $+0.186$ on average. The $\Delta\text{FPR} = 0$ guarantee holds for single-subject contamination ($k \leq 5$, $\Delta\text{FPR} < 0.007$). Under broader contamination spanning $k = 10$ subjects (18% of domains), ΔFPR rises to 0.047 ± 0.022; at $k = 20$, to 0.069 ± 0.018 (Table 6 and fig. 3). Even worst-case multi-subject inflation (∼7 pp) cannot account for the observed 31% C% rate, which exceeds the 0.18% random baseline by $>170\times$. Under DEFF adjustment, worst-case

Table 3: Temporal DIF across six METABENCH benchmarks. C% = ETS C items ($|\Delta_{\mathrm{MH}}| \geq 1.5$) between 2023 and 2024 cohorts; ρ/τ = Spearman/Kendall correlation between full and stable-item rankings; Flip = pairwise rank-flip rate; Ratio = C%/Rand. C%, with baseline varying by benchmark. All six show substantial DIF (45–174× above random baseline), with knowledge/reasoning benchmarks exhibiting higher C% than procedural ones.

Benchmark	Items	C%	ρ	τ	Flip%	Rand. C%	Ratio
MMLU	12,508	31.3	0.9965	0.9600	2.00	0.18	174×
WinoGrande	1,267	30.3	0.9943	0.9432	2.84	0.32	95×
TruthfulQA	817	29.7	0.9971	0.9586	2.07	0.37	80×
ARC	1,172	26.9	0.9962	0.9538	2.31	0.60	45×
GSM8K	1,319	18.9	0.9993	0.9843	0.78	0.30	63×
HellaSwag	10,042	17.9	0.9986	0.9722	1.39	0.28	64×

ΔFPR at k=20 (0.069) represents up to 50% of the 13.5% lower bound, underscoring $k \leq 5$ in practice. Difficulty-dependent injection that captures the observed $\rho = -0.597$ gradient remains future work.

4.6 Downstream Impact

Stable-item ranks are preserved across all six benchmarks ($\rho = 0.997$, $\tau = 0.958$; flip rate $\leq 2.8\%$; Table 3). DIF-flagged items concentrate discriminative signal ($z = -8$ to -49 vs. random removal; Figure 4). Among the 200 most displaced models (top 3.8%), 33.6% of pairwise rankings reverse, concentrating among closely-ranked pairs (35.2% at Δrank < 50 vs. 0.0% above 1000; Section J). Removing C-flagged items disproportionately disadvantages 2024 models (cohort gap = 96.2 positions, 40× random; $p < 0.001$).

4.7 Cross-Benchmark Validation

Temporal DIF extends across all six METABENCH benchmarks (Table 3): C% ranges from 17.9% (HellaSwag (Zellers et al., 2019)) to 31.3% (MMLU), with knowledge/reasoning benchmarks consistently higher than procedural ones (C% = 26.9–31.3% vs. 17.9–18.9%). These differences may partly reflect item difficulty distributions; we treat them as descriptive.

Semi-synthetic injection on GSM8K confirms detection power (ΔTPR = +0.514; Section E); GSM1K analysis ($r = -0.478$, $p = 0.005$; Figure 5) suggests DIF captures beyond memorization.

5 Discussion

5.1 What Drives Temporal DIF?

Temporal DIF detects *that* items function differently across cohorts, not *why*. Three non-exclusive mechanisms may drive the observed ≈27–29% rate: (1) differential contamination; (2) capability evolution; (3) training methodology shifts. The ETS baseline of 5–15% C% (Clauser and Mazor, 1998) is for concurrent demographic groups; the

are near-independent of DIF flags (Jaccard = 0.206; Section 4.3).

The aggregate difficulty–direction interaction ($\rho = -0.597$; Section 4.2) is suggestive: hardest-decile DIF-C items are 88.7% focal-favoring, easiest-decile 96.2% reference-favoring ($\rho = -0.447$ at $[0.2, 0.8]$; $p < 10^{-100}$, ruling out ceiling effects); however, observational DIF alone cannot separate contamination from capability evolution (Suk and Lyu, 2026). The aggregate $\rho \approx 0.997$ masks subject-level instability (22/57 below 0.95; five below 0.90), motivating targeted replacement (Zhao et al., 2025; Habba et al., 2026).

Stratifying by model type reveals distinct mechanisms: base models (N=1,409) are 60.1% reference-favoring with a strong difficulty gradient ($\rho = -0.570$), consistent with differential memorization decay, while instruct models (N=478) are 65.6% focal-favoring with a near-zero gradient ($\rho = -0.079$; $\chi^2 = 562.7$, $p < 10^{-124}$), consistent with uniform alignment improvements. The aggregate gradient is driven by base models; the instruct flat gradient is compatible with either capability gains or difficulty-independent memorization. This decomposition provides positive evidence for mechanism separation, complementing the exclusion arguments above.

5.2 Recommended DIF Audit Protocol

Item-level properties do not predict DIF susceptibility (AUROC = 0.60; Section G), so monitoring cannot be replaced by pre-screening. We recommend MH-DIF audits whenever ≥500 models accumulate (ensuring ≥80 per stratum at K=5; Section 4.1), with three action levels (calibrate via ρ-degradation; Figure 6): *Monitor* (C% < 15%), *Flag* (15–25%: dual rankings), *Act* (≥ 25%: stable-item rankings). Difficulty–direction and model-type signals (Section 5.1) inform triage; DIF removal is the least disruptive strategy (Figure 4), motivating equating (Kolen and Brennan, 2014; Habba et al., 2026). Subject-level monitoring is essential (22/57 below $\rho = 0.95$).

6 Conclusion

Across six METABENCH benchmarks, 17.9–31.3% of items exhibit temporal DIF between 2023 and 2024 cohorts (≈27–29% for MMLU after non-independence corrections; conservative bound: 13.5%). Web-overlap labels are near-independent of DIF flags (Jaccard = 0.206); rankings are preserved ($\rho \approx 0.997$), motivating routine DIF audits.

Limitations

Three primary limitations apply. (1) The semi-synthetic framework uses uniform binary response flipping, a worst-case injection model; reported ΔTPR values are upper-bound estimates, and detection power under realistic contamination patterns (partial memorization, format-specific advantages) is untested. More fundamentally, DIF detects *that* items function differently but cannot determine *why*. (2) The MH χ^2 test assumes independent examinees; LLM model families violate this assumption. Three convergent corrections ($\approx$27–29%) and family-level cluster bootstrap (95% CI: $[26.7\%, 53.2\%]$) bound the impact, but the extreme relatedness of some families (e.g., hundreds of Llama-2 fine-tunes) exceeds the clustering levels studied in educational settings (French and Finch, 2010). (3) Cross-subject matching assumes contamination affects a minority of domains; ΔFPR rises to 0.069 when contamination spans $\geq$20 subjects. Temporal cohorts defined by calendar year provide coarse granularity, and generalizability to open-ended or partial-credit benchmarks is untested. Multidimensional DIF (Ackerman, 1992) may arise when MMLU's 57 subjects span distinct latent dimensions; cross-subject matching mitigates between-domain multidimensionality but not within-domain violations. IRT parameters are used only for auxiliary power analysis; all core DIF results are nonparametric.

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A Method Details

A.1 Diagnostics

Three diagnostic procedures complement the primary DIF analysis.

Local independence (Yen's Q3). The MH procedure assumes that item responses are conditionally independent given θ. Yen's Q3 statistic measures residual correlations between item pairs after conditioning on the latent trait. We compute Q3 for all within-subject and cross-subject item pairs, with $|Q3| > 0.20$ as the threshold for local dependence. Elevated within-subject dependence would indicate that items within the same subject share variance beyond what θ captures, which could inflate within-subject DIF rates.

Domain-specificity. We test whether temporal DIF concentrates in domains where models improved most. At the ecological level, we compute Spearman ρ between subject-level C% and subject-level accuracy improvement from 2023 to 2024. At the item level, logistic regression predicts item-level DIF C classification from domain-level accuracy change, with and without item property controls (difficulty, discrimination). This decomposition quantifies how much of the observed instability is attributable to aggregate domain-level gains versus item-specific behavioral change.

Instability decomposition. When domain accuracy improvement is included alongside item-level controls, we compute the counterfactual C% that would remain after removing domain-level effects. The gap between observed and counterfactual C% measures the contribution of domain-level improvement to the total instability.

A.2 Statistical Considerations

Multiple testing. The ETS C classification imposes a dual threshold (effect size $|\Delta_{\mathrm{MH}}| \geq 1.5$ and significance $p < 0.05$), which provides implicit control over false positives. As a robustness check, we apply Benjamini-Hochberg FDR correction at $q = 0.05$. The random-split sanity check provides an empirical estimate of the false positive rate under the null hypothesis.

Threshold sensitivity. We report results across a range of DIF thresholds ($|\Delta_{\mathrm{MH}}| \in \{1.0, 1.5, 2.0, 2.5\}$) to verify that qualitative conclusions do not depend on the specific ETS C cutoff. The standard threshold of $|\Delta_{\mathrm{MH}}| \geq 1.5$ corresponds to the ETS C definition used throughout the paper.

IRT model fit. For transparency, we report that the MMLU response matrix shows poor fit to a unidimensional 2PL model (expected for a 57-subject multidimensional benchmark). This does not affect MH-DIF validity, which relies only on total score stratification rather than parametric IRT assumptions.

Architecture heterogeneity. The METABENCH MMLU population consists of 99.98% decoder-only models (5,226 of 5,227). This near-complete architectural homogeneity eliminates architecture as a confound for DIF analysis. We verify this by excluding the single encoder-decoder model and confirming identical results.

Power sensitivity to item discrimination. The Monte Carlo power analysis in Section 4.1 uses average 2PL parameters ($\bar{a} = 1.2$, $\bar{b} = -0.84$). Table 4 reports TPR as a function of discrimination a (other parameters at their means, $N = 80$ per cohort, $|\Delta_b| = 1.4$, 1,000 replications). TPR increases monotonically from 0.40 ($a = 0.5$) to 0.95 ($a = 2.0$); FPR remains below 0.04 throughout. The MMLU item pool shows substantial heterogeneity: a ranges from 0.006 to 2.657 (SD = 0.54, IQR = $[0.75, 1.64]$). The aggregate TPR of 0.68 from Phase 2 reflects this full distribution, while TPR at mean $a \approx 1.2$ is 0.87—the gap is consistent with Jensen's inequality applied to the concave TPR(a) function. Although 2PL fit is poor for MMLU's multidimensional structure, the power analysis serves as one leg of a triangulation: parametric estimates (TPR = 0.68–0.80) are validated by empirical semi-synthetic injection ($\Delta\mathrm{TPR} = +0.508$ at $p = 0.5$; Section 4.5), confirming adequate detection power regardless of parametric assumptions.

Table 4: Power sensitivity to item discrimination. TPR = true positive rate for ETS C detection ($|\Delta_{\mathrm{MH}}| \geq 1.5$) at $N = 80$ per cohort, $|\Delta_b| = 1.4$, 1,000 replications.

a	TPR	95% CI	FPR
0.5	0.398	[0.20, 0.60]	0.033
0.8	0.683	[0.45, 0.90]	0.036
1.0	0.808	[0.60, 0.95]	0.038
1.2 ($\approx \bar{a}$)	0.874	[0.70, 1.00]	0.038
1.5	0.930	[0.80, 1.00]	0.038
2.0	0.949	[0.80, 1.00]	0.037

Nearest-neighbor matching. The 1:1 nearest-neighbor matching selects, for each focal-cohort model, the reference-cohort model with the closest total score (no caliper). Balance is verified by Cohen's $d = -0.17$ on matched pairs ($N = 638$ per group), confirming minimal residual ability confounding.

Logistic regression specification. The Simpson's paradox analysis (Section 4.2) uses binary logistic regression with DIF-C classification as the dependent variable, domain-level accuracy change (2024 minus 2023 mean accuracy) as the primary predictor, and item difficulty (proportion correct) and IRT discrimination (point-biserial correlation) as controls. The interaction between domain accuracy change and item difficulty is included. Non-uniform DIF detection (Section 4.4) uses logistic regression with a group $\times$ ability interaction term; a significant interaction ($p < 0.05$) indicates non-uniform DIF.

B Structural Analysis Details

Two structural controls and a sensitivity check fail to reduce the temporal C% below 31%, revealing a structural property of the data rather than a methodological limitation. Matched-ability quintile analysis restricts comparisons to models of similar total score and yields enrichment ratios of 0.94–1.04, with temporal C% unchanged at approximately 0.31. Within-subject DIF analysis on 5 representative subjects, intended to isolate subject-specific effects, finds mean C% = 38.1%, exceeding the 25% threshold that would indicate viable bifactor structure. As a sensitivity check, IRT θ-conditioning with 20 ability strata achieves $r(\theta, \text{total score}) = 0.9986$ and C% = 0.312, confirming that the external total score already captures nearly all ability variance relevant to DIF computation. Table 2 summarizes these results.

The temporal FPR reflects genuine item-level instability that persists regardless of how ability is measured or conditioned. Yen's Q3 diagnostic reveals mild local dependence within subjects (11.3% of within-subject item pairs exceed $|Q3| > 0.20$, vs. 2.2% cross-subject; 5$\times$ ratio, $p = 2.3 \times 10^{-9}$), which contributes to elevated within-subject DIF rates but is not the primary driver: the cross-subject matching protocol addresses this bias, and temporal C% persists under external matching (full Q3 distributions available in the supplementary material).

Table 5: Domain-level Cohen's d between 2023 and 2024 cohorts. The narrow range (0.12) indicates nearly uniform improvement across domains, supporting total-score matching adequacy for DIF analysis (Section 3).

Domain	Cohen's d
STEM	0.261
Humanities	0.159
Social Science	0.147
Other	0.146

Table 6: Multi-subject injection robustness. k = number of simultaneously contaminated subjects (out of 57); values are mean $\pm$ SD across 5 random subject draws at exploitation rate $p = 0.5$.

k	ΔFPR	ΔTPR
1	$0.000_{\pm.000}$	$0.371_{\pm.048}$
3	$0.003_{\pm.002}$	$0.385_{\pm.031}$
5	$0.007_{\pm.003}$	$0.392_{\pm.027}$
10	$0.047_{\pm.022}$	$0.401_{\pm.024}$
20	$0.069_{\pm.018}$	$0.410_{\pm.020}$

Table 7: Leave-one-family-out (LOFO) robustness on MMLU. Removing non-dominant families ($\leq$15% of their cohort) changes C% by at most 2.93 pp. Larger shifts from removing Mistral or Llama-3 are driven almost entirely by family composition, not statistical power loss: Monte Carlo random-subsample calibration (200 replicates per scenario, matching the reduced sample size) shows the power effect is $\leq$1 pp, while the composition effect accounts for 95–113% of the observed C% drop (Mistral: -8.44 pp composition vs. -0.41 pp power; Llama-3: -8.17 pp vs. $+0.91$ pp; observed C% falls below the 95% random-subsample range in both cases). Stratification uses total test score for computational efficiency; cross-subject matching yields identical baseline C%.

Family	N removed	Cohort share	C% (LOFO)
Full sample	—	—	31.32
Mistral	547	51% ref	22.47
Llama-3	315	39% foc	24.06
Mixtral	120	15% foc	34.26
Llama-2	161	15% ref	31.48
Solar	91	11% foc	31.01

C Design Effect Estimation

Model families (e.g., Mistral, Llama-3) induce intra-class correlation (ICC) that inflates Mantel-Haenszel χ^2 statistics beyond the nominal χ^2_1 reference distribution. We estimate the design effect (DEFF) per item following Rao and Scott (1992): $\mathrm{DEFF}_i = 1 + (\bar{m} - 1) \cdot \mathrm{ICC}_i$, where $\bar{m}$ is the mean family size and ICC_i is the one-way random-effects ICC for item i across 30 families (restricted to ref+foc cohort; 1,884 models with family$\geq$2).

Two estimates of $\bar{m}$ bracket the adjustment: (1) *Simple mean* ($\bar{m} = 62.8$), treating all families equally, yields DEFF $= 3.87$ and adjusted C% $=$ 29.2%; (2) *Weighted mean* ($\bar{m} = \sum n_f^2 / \sum n_f =$ 254.5; Kish, 1965), reflecting the effective cluster size experienced by a randomly sampled model, yields DEFF $= 12.79$ and adjusted C% $= 13.5\%$. Mean ICC $= 0.047$ (median 0.041, Q3 $= 0.061$, max $= 0.219$; 4.8% of items exceed 0.10). After dividing each item's χ^2_{MH} by its item-specific DEFF and re-applying ETS thresholds, 13.5–29.2% of items retain Category C classification depending on the $\bar{m}$ variant—both far exceeding the 0.18% random-split baseline (75–162$\times$).

D Uncertainty Quantification Comparison

The cluster bootstrap upper bound (53.2%) is driven by extreme bootstrap resamples that over-

Table 8: Comparison of uncertainty quantification methods for the 31.3% C% estimate on MMLU.

Method	Key assumption	Result
Model-level bootstrap	Models independent	95% CI [29.4%, 38.1%]
Family-level cluster bootstrap	Family is clustering unit	95% CI [26.7%, 53.2%]
DEFF (simple $\bar{m}$)	Equal-weight families	Adjusted C% = 29.2%
DEFF (weighted $\bar{m}$)	Size-proportional weighting	Adjusted C% = 13.5%
Graduated de-dup. (K=20)	Remove within-family duplicates	C% = 19.7% $\pm$ 0.9%

represent Mistral ($n = 547$) or Llama-3 ($n = 315$), producing atypical cohort compositions. Two independent methods—cluster bootstrap lower bound (26.7%) and simple-mean DEFF adjustment (29.2%)—converge on $\sim$27–29%, establishing a robust corrected estimate that remains $>$150$\times$ the random-split baseline (0.18%). Graduated deduplication at K=20 yields C% $= 19.7\% \pm 0.9\%$ (random member selection, 5 draws), lower than the corrected estimate due to reduced statistical power ($N \approx 450$ vs. 1,887), but within-K variance $<$1.5 pp at all K confirms the signal is power-driven rather than redundancy-inflated (Section I). The weighted-mean DEFF (13.5%) represents a conservative extreme; even this lower bound exceeds the baseline by 75$\times$.

E Cross-Benchmark Comparison

Table 9: Cross-benchmark comparison of DIF analysis on MMLU and GSM8K. Both benchmarks exhibit temporal measurement instability with validated semi-synthetic detection power. Standard and external matching use different exploitation rates (p) because each benchmark's optimal operating point differs; external matching results are compared at $p = 0.5$ for both. Subscript $\pm$ values denote standard deviations across 5 random injection seeds.

Metric	MMLU	GSM8K
Items	12,508	1,319
Models	5,227	6,702
Temporal C%	31.3%	18.9%
Sanity C%	0.18%	0.3%
ΔTPR (p=0.7, standard)	–	$+0.818_{\pm.006}$
ΔTPR (p=0.5, external)	$+0.508_{\pm.027}$	$+0.514_{\pm.028}$

F Additional Figures

G Item-Property Predictive Analysis

To test whether DIF susceptibility is a fixed item property, we trained logistic regression and LightGBM classifiers to predict ETS C classification from 17 leak-free features: ref-cohort difficulty and discrimination, 11 text features (question/option length, vocabulary metrics), subject metadata, and

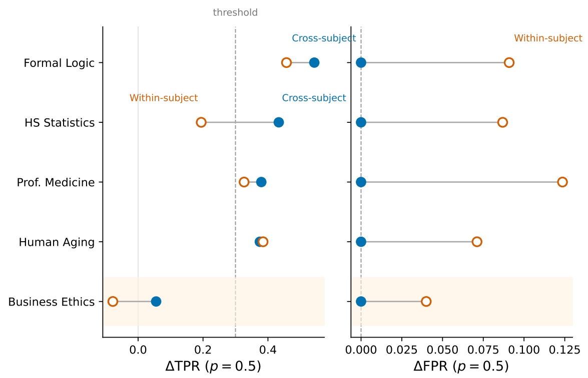

Figure 3: Cross-subject vs. within-subject matching for semi-synthetic DIF detection. Cross-subject matching eliminates FPR inflation (ΔFPR $= 0.000$) while preserving detection power (ΔTPR > 0.3 for 4/5 subjects at $p = 0.5$). Within-subject matching inflates FPR by $+0.186$ on average.

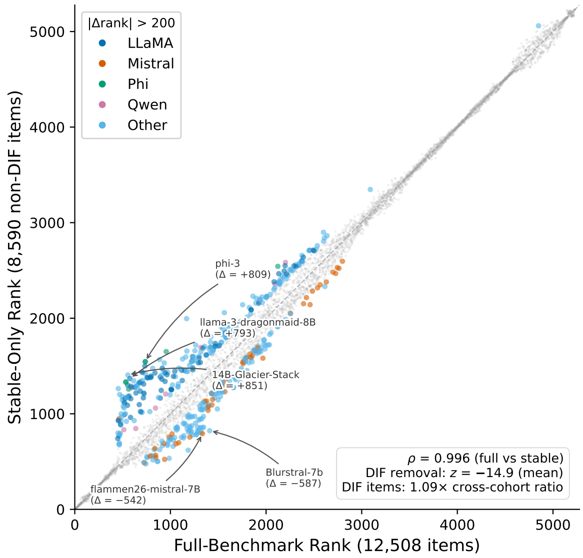

Figure 4: Downstream impact of DIF-based item filtering on model rankings. Despite $\rho \approx 0.997$ overall, individual model families show systematic rank shifts when evaluated on stable items only.

web-overlap labels (Li et al., 2024b). Features were computed exclusively from the reference (2023) cohort to prevent information leakage from the focal cohort. Five-fold subject-stratified cross-validation (no within-subject train/test overlap) yielded AUROC $= 0.603 \pm 0.020$ (LR) and 0.599 ± 0.026 (LightGBM)—near chance. Web-overlap labels contributed negligibly (SHAP < 0.015), independently confirming the DIF–contamination orthogonality reported in Section 4.3. These results indicate that temporal DIF reflects emergent cohort–item interactions rather than fixed item defects, motivating post-hoc monitoring over pre-screening.

H ICC by Family Size

Within-family similarity (mean pairwise Pearson r over 12,508-item response vectors) shows no dependence on family size (Spearman $\rho = -0.009$, $p = 0.96$; 30 families with ≥ 2 members, 1,884 models). Large families (Mistral, $m = 547$, $\bar{r} = 0.47$; Llama-3, $m = 315$, $\bar{r} = 0.53$) are no more homogeneous than small or mid-sized families. Of the 5,227 models, 3,343 lack family assignments; these are retained in DIF analysis but excluded from similarity estimation. Within the temporal DIF cohorts ($N_{\text{ref}} + N_{\text{foc}} = 1{,}887$), 1,884 (99.8%) belong to identified families with ≥ 2 members, so DEFF estimates are computed over nearly the full cohort. This indicates that the weighted-mean DEFF variant's effective cluster size ($\bar{m} = 254$) reflects an arithmetic artifact of mega-family dominance rather than genuine within-family homogeneity.

I Graduated De-Duplication Sensitivity

For each value of K (max models retained per family), K members are sampled uniformly at random; remaining family members are excluded from both reference and focal cohorts, and MH-DIF is recomputed on the reduced sample (5 independent draws per K; $K =$ all is deterministic).

C% increases monotonically from 0.3% ($K = 1$) to 31.3% ($K =$ all). At each fixed K, variance across random member draws is negligible (SD $<$1.5 pp), confirming that the signal scales with statistical power rather than within-family redundancy. At K=20, C% $= 19.7\% \pm 0.9\%$ ($N \approx 450$); the gap from the full-sample 31.3% reflects reduced power, not reduced bias.

J Rank-Reversal by Distance

Pairwise rank reversals between full-item and stable-item rankings concentrate among closely-ranked models. At Δrank < 50 (mean Δacc $= 0.003$, effectively statistical ties), the reversal rate is 35.2%. It drops to 15.1% for Δrank $\in [50, 200)$, 4.0% for $[200, 500)$, 0.6% for $[500, 1000)$, and 0.0% above 1000. Broad leaderboard structure is fully preserved; instability affects only models within each other's noise margin.

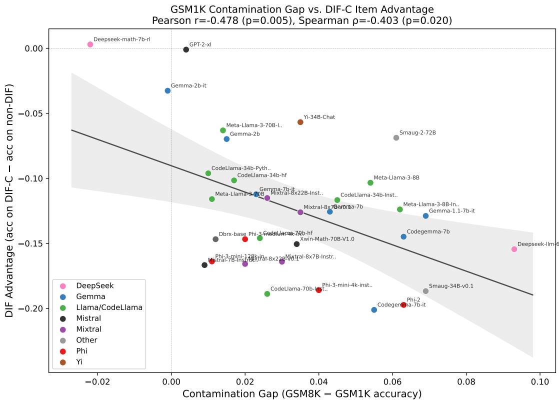

Figure 5: Exploratory: GSM1K contamination gap vs. DIF ranking advantage ($N = 33$; small sample, results should be interpreted cautiously). The negative correlation ($r = -0.478$, $p = 0.005$) indicates that models with larger contamination gaps benefit less from DIF-flagged items. Three of 33 models exceed the Cook's D threshold ($4/N = 0.12$); leave-one-out analysis confirms all individual-removal correlations remain significant ($r \in [-0.54, -0.37]$, all $p < 0.05$), and removing all three outliers yields $r = -0.38$ ($p = 0.036$, $N = 30$).

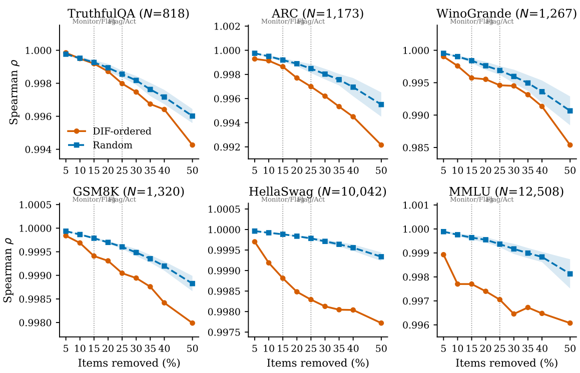

Figure 6: Ranking degradation under DIF-ordered vs. random item removal across six benchmarks. DIF-ordered removal (orange) consistently degrades ρ faster than random removal (blue band = mean $\pm$ 1 SD, 20 seeds). Vertical dashed lines mark the Monitor/Flag (15%) and Flag/Act (25%) thresholds from Section 5.2.